COMP 9200
Applied Research Methods

Data in AI and machine learning

Overview

- AI vs ML vs algorithm
- ML algorithms and design
- Data issues
- Analyzing ML results
- Planning in ML research

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AI and Machine Learning

- Artificial Intelligence (AI): Simulating human intelligence in machines
- Machine Learning (ML): Subset of AI that learns from data
- Types of ML algorithms: Supervised, Unsupervised, Reinforcement Learning
- Data: Training, Testing, Validation

Machine learning algorithms

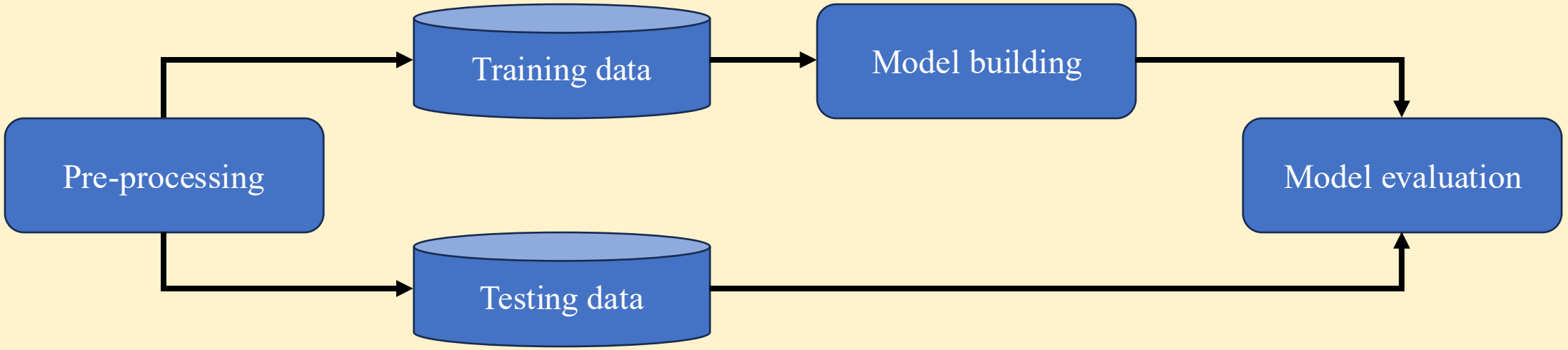
1. Define the goal and available data
2. Choose problem type to determine algorithm type
3. Select specific algorithm
   - Consider the nature of the problem (classification, regression, clustering).
   - Evaluate data size, quality, and structure.
   - Consider interpretability and computational efficiency.
4. Prepare and split data
5. Train, test and validate ML
6. Analyze results

Select / design ML algorithms

Question	Problem type	Algorithm Type
Do you need to predict a category or label? (e.g., spam or not spam, cat or dog)	Classification	Supervised Learning
Do you need to predict a continuous value? (e.g., house price, temperature, stock value)	Regression	Supervised Learning
Do you need to discover inherent groupings or structure in data without predefined labels? (e.g., market segmentation, grouping similar docs)	Clustering	Unsupervised Learning
Do you need to reduce the number of features while preserving information? (e.g., data compression, noise reduction)	Dimensionality Reduction	Unsupervised Learning
Do you need a system to make a sequence of decisions or maximize a reward through interaction with an environment? (e.g., robotics, game playing)	Reinforcement Learning	Reinforcement Learning

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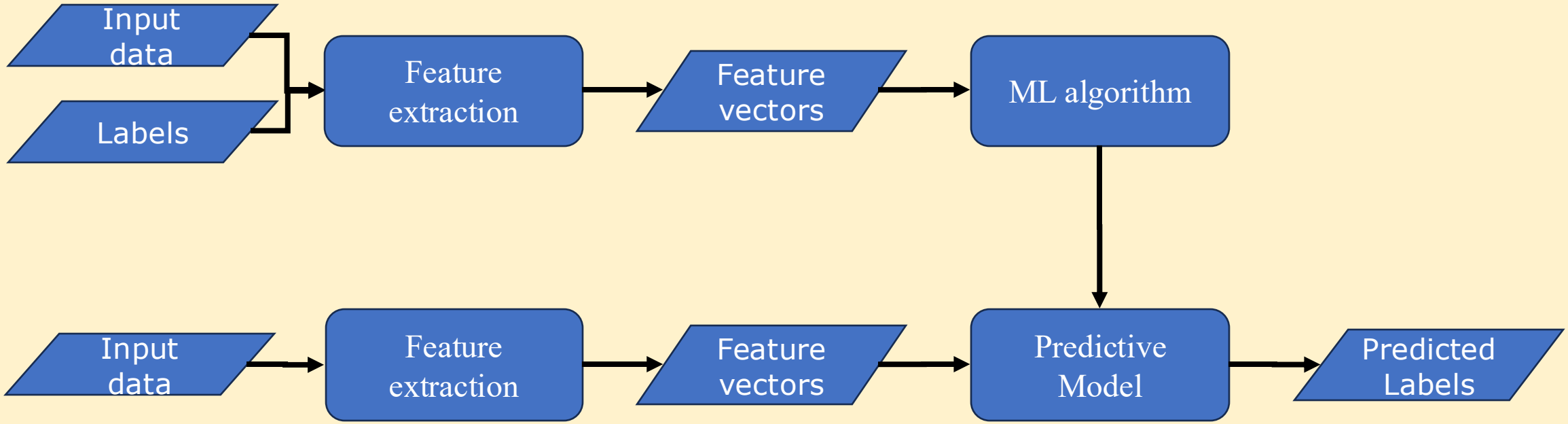
Select / design ML algorithms



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Select / design ML algorithms

Labelled data available – Supervised learning



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ML algorithms

Supervised Learning

- Model trained on labeled dataset (input-output pairs)
- Goal to learn mapping from inputs to outputs
- Classification: predict category or label
- Regression: predict continuous value
- Real-World Example:
  - Email Spam Detection: Classify emails as “spam” or “not spam” based on labeled examples.
  - Stock predictor: Forecast stock performance

ML algorithms

Supervised Learning

- Pros:
  - High accuracy when sufficient labeled data is available.
  - Well-understood algorithms and metrics.
  - Works well for prediction tasks.
- Cons:
  - Requires large labeled datasets (expensive to create).
  - Can overfit if not regularized.
  - Performance depends heavily on data quality.

ML algorithms

Semi-Supervised Learning

- Uses small amount of labeled data and large amount of unlabeled data to improve learning

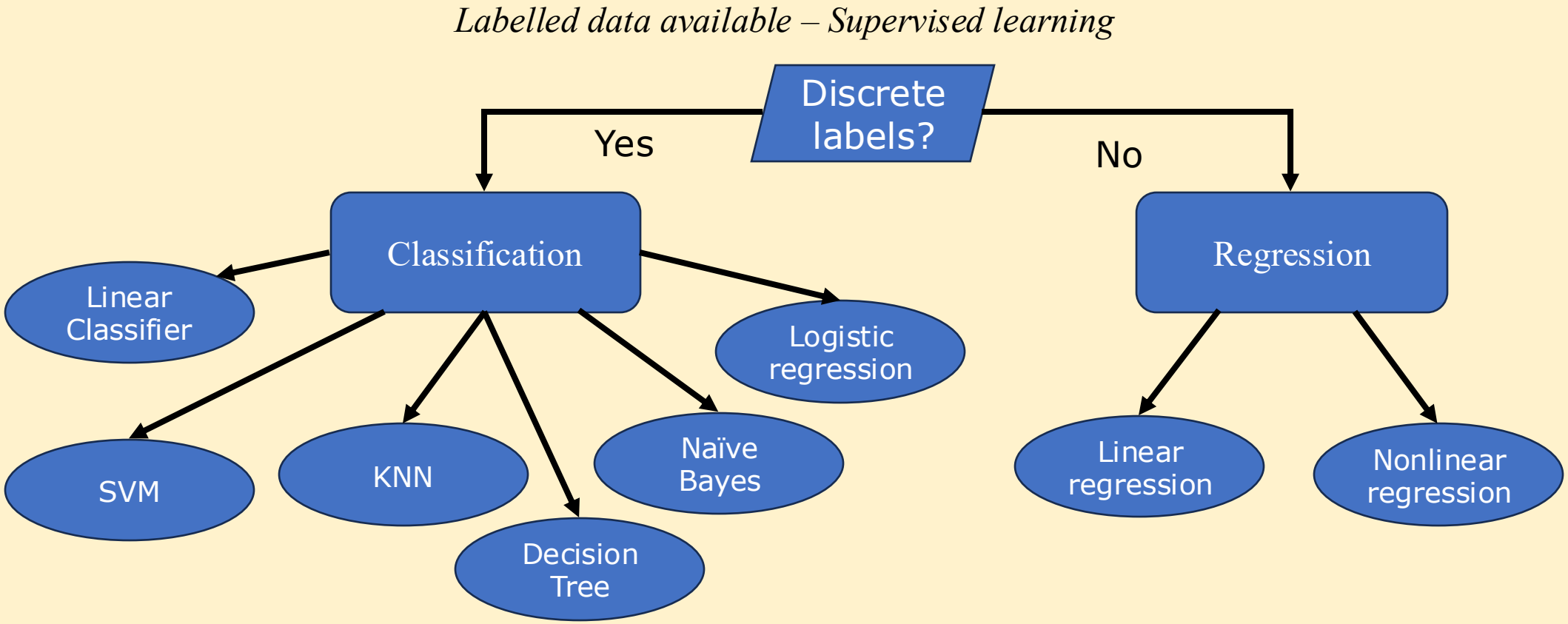
- Real-World Example:
  - Medical Image Classification: Few labeled scans with many unlabeled ones.

ML algorithms

Semi-Supervised Learning

- Pros:
  - Reduces labeling cost.
  - Often achieves better performance than unsupervised learning.
  - Useful when labeled data is scarce.
- Cons:
  - Assumes unlabeled data follows the same distribution as labeled data.
  - More complex than purely supervised or unsupervised methods.
  - Sensitive to incorrect labels.

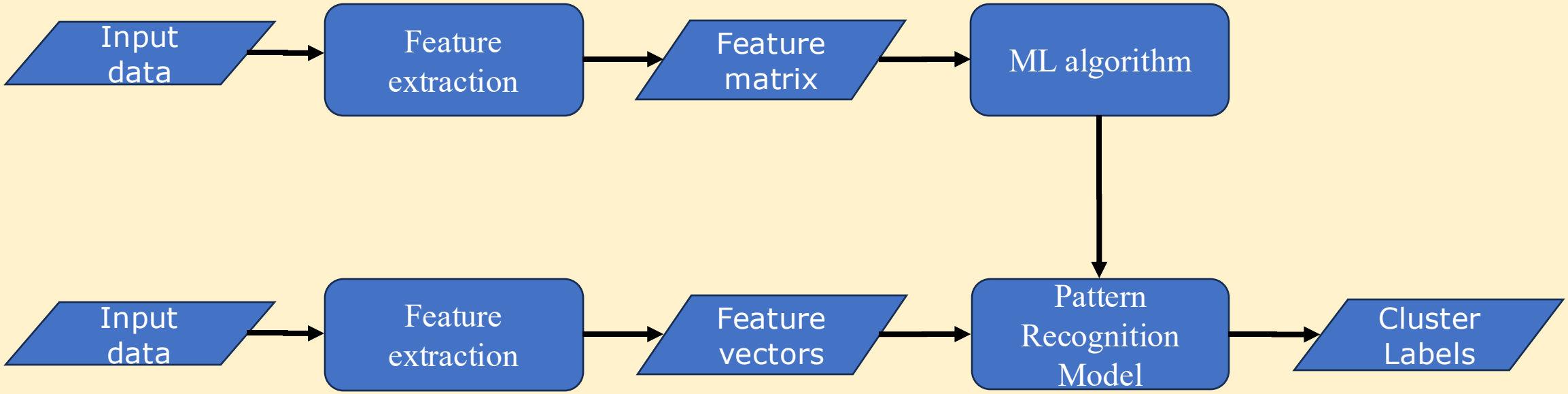
Select / design ML algorithms



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Select / design ML algorithms

Labelled data not available – Unsupervised learning



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ML algorithms

Unsupervised Learning

- Model trained on unlabeled data
- Goal to find hidden patterns or groupings
- Clustering: discover inherent groupings or structure in data without predefined labels
- Dimensionality reduction: reduce number of features while preserving information
- Real-World Example:
  - Customer Segmentation: Group customers based on purchasing behavior without predefined labels.
  - Noise reduction: Reduce noise in data

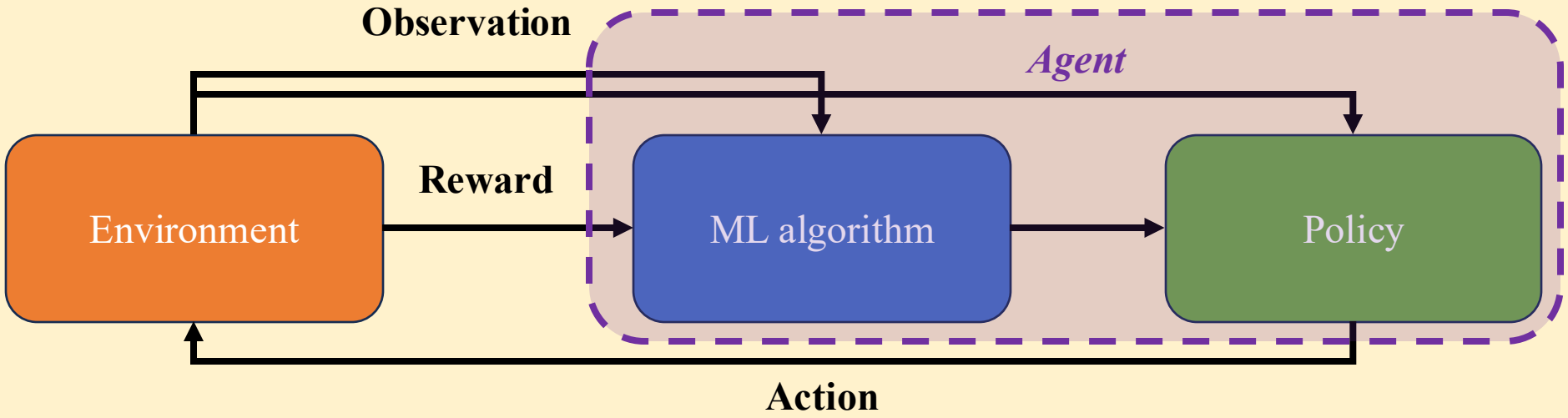
ML algorithms

Unsupervised Learning

- Pros:
  - No need for labeled data (cheaper).
  - Useful for exploratory analysis and pattern discovery.
  - Can reveal hidden structures in data.
- Cons:
  - Hard to evaluate results objectively.
  - May produce clusters that are not meaningful.
  - Sensitive to parameter choices and scaling.

Select / design ML algorithms

Reinforcement learning – learn from environment



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ML algorithms

Reinforcement Learning

- Agent learns by interacting with an environment, receiving rewards or penalties for actions
- System to make sequence of decisions or maximize a reward through interaction with an environment?
- Real-World Example:
  - Autonomous Driving: A car learns to navigate by maximizing safety and efficiency rewards.

ML algorithms

Reinforcement Learning

- Pros:
  - Handles sequential decision-making problems.
  - Learns optimal strategies through trial and error.
  - Useful for dynamic environments.
- Cons:
  - Requires many interactions (computationally expensive).
  - Can be unstable or fail to converge.
  - Reward design is challenging.

Data issues

Datasets and splitting

- Training set: used to train the model
- Validation set: used to tune hyperparameters
- Test set: used to evaluate final model performance

Improper splitting can lead to biased or misleading results

Data issues

- Sources of bias:
  - Sampling
  - Labeling
  - Historical data
  - Algorithm design
- Techniques to minimize bias:
  - Data augmentation: diverse and representative datasets
  - Fairness-aware algorithms
  - Regular audits and evaluations

Other issues to consider

- Overfitting and underfitting
- Model complexity and interpretability
- Computational resources and scalability
- Feature selection
- Hyperparameter tuning
- Ethical considerations and transparency

Analyzing ML results

- Use metrics: accuracy, precision, recall, F1-score, ROC-AUC
- Visualize results with confusion matrices, ROC curves
- Interpret results in context of research goals and limitations
- Validate findings with domain experts

Planning in ML research

- Define clear research questions and hypotheses
- Select and design ML algorithm with justification
- Plan data collection, splitting and preprocessing steps
- Examine closely bias and ethics
- Ensure reproducibility and transparency in methods
- Good detailed documentation to maximize reproducibility and transparency in methods and validation of findings

Review

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